SERVER:
COMPOSABLE MEMORY
SYSTEMS (CMS)



Beyond DRAM: Persistent Memory via Compute Interconnects for AI

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AI Storage Demand: The KV Cache Bottleneck

Training: High-throughput storage

Large-scale data loading and checkpoint writes

Inference: KV cache expansion

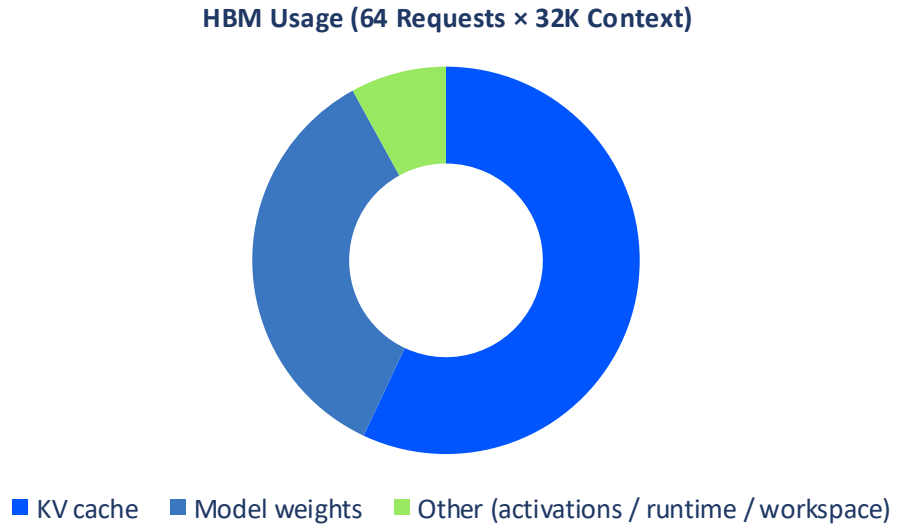
Long contexts and high concurrency increase memory demand

Agentic AI: Longer data residency

Multi-turn contexts and tool-call chains keep accumulating KV

KV cache uses 57% of HBM in the example workload

64 concurrent requests × 32K context; larger than the model-weight share



	Model weights	KV (64 × 32K)
Qwen3-235B	235 GB	~384 GB
DeepSeek-V3	671 GB	~141 GB

Storage Media Tiering for AI Workloads

No single medium meets all capacity, performance and cost needs

HBM / DRAM

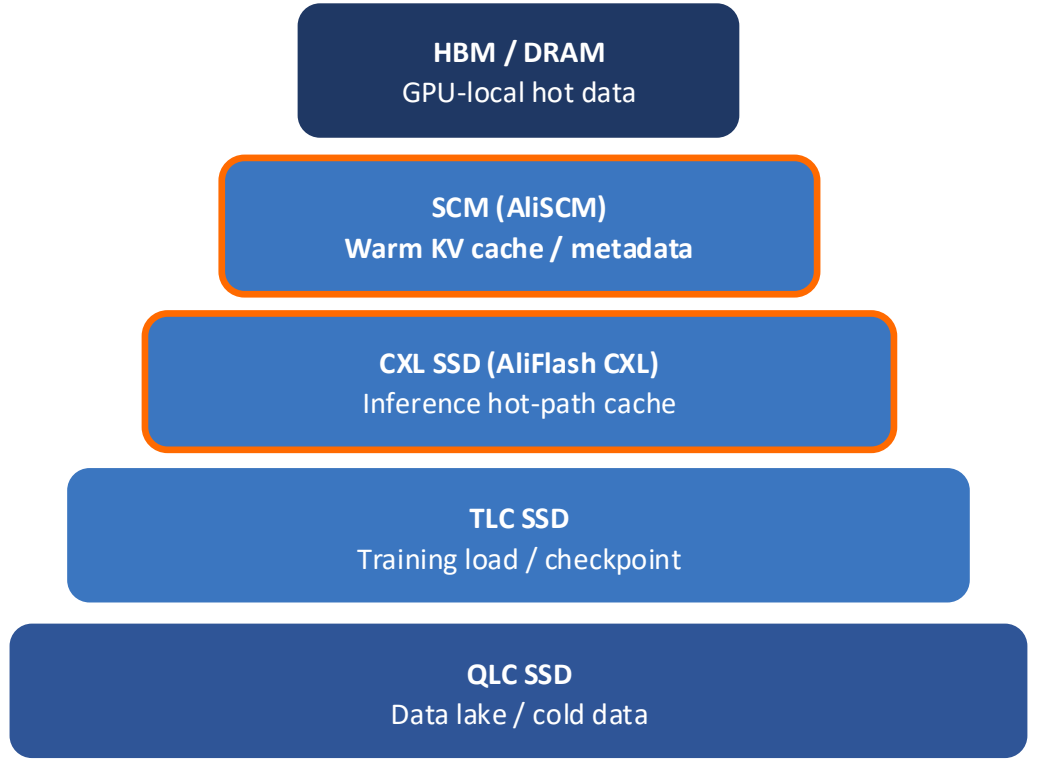
Keep the active GPU working set

AliSCM and AliFlash CXL SSD

Expand external capacity with PCM and NAND tiers

TLC / QLC SSD

Serve training loads, checkpoints and cold data



Top: fastest · smallest · highest cost Bottom: slowest · largest · lowest cost per GB

AliSCM: Alibaba Cloud's Persistent Memory

In volume production and deployment

CXL 2.0 Type-3 · E3.S

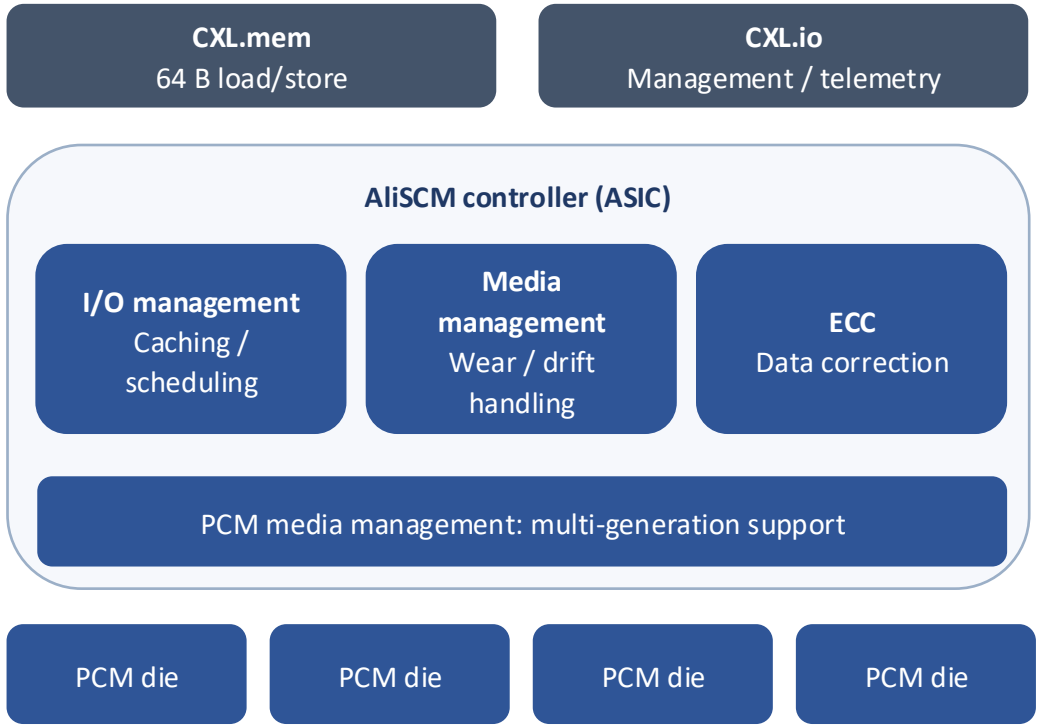
256 GB–1 TB per module

Byte-addressable with power-loss persistence



Key Measured Metrics

Metric	AliSCM	Optane DIMM
Addressing granularity	128 B	256 B
Read latency	350 ns	300 ns
Read bandwidth	20 GB/s	8.1 GB/s
Write bandwidth	6.5 GB/s	3.15 GB/s



AliFlash CXL: Gen7 High-IOPS Storage

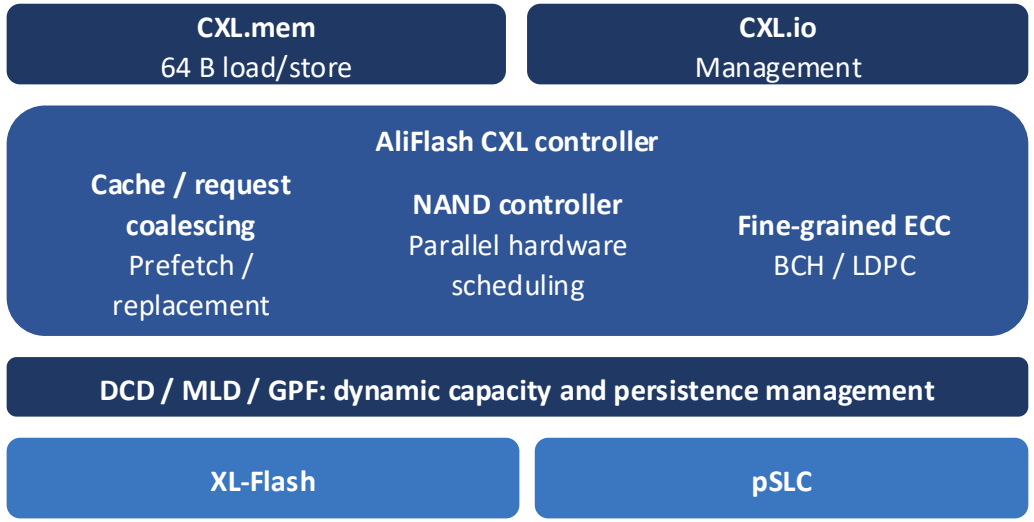
High-IOPS capacity for fine-grained GPU KV cache access

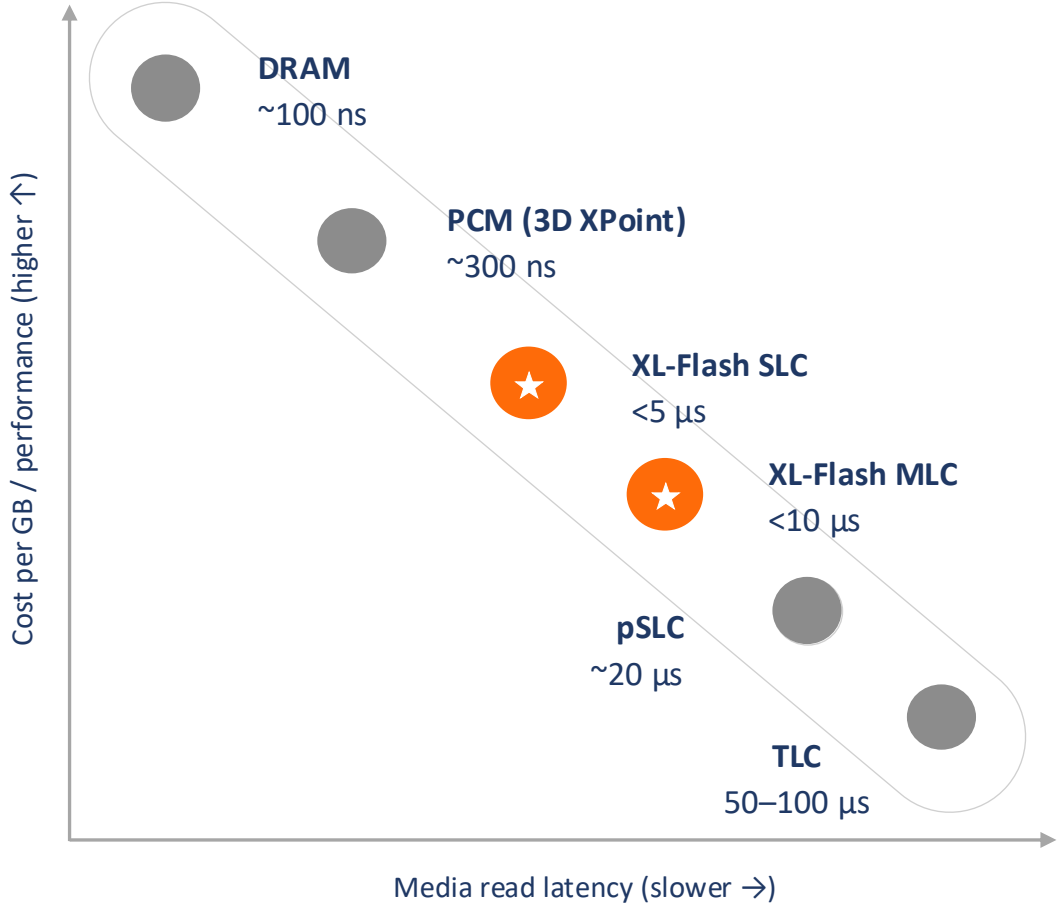
Gen6 / CXL 3.2: E3.S, TB-class capacity tier

Gen7 / CXL 4.0: 50–100M IOPS target per device

512 B random reads: 51.2 GB/s useful bandwidth at 100M IOPS

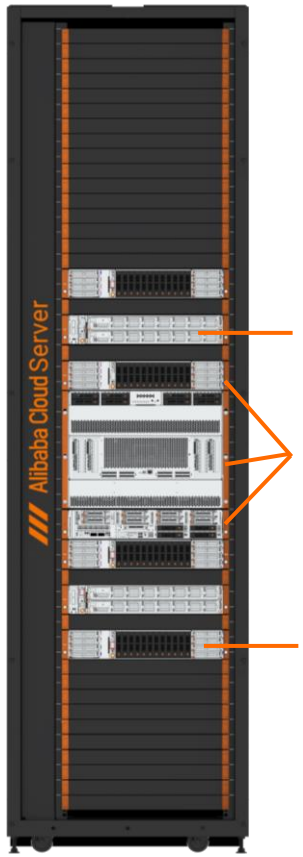
PCIe Gen7 ×4 · CXL.mem load/store access





CXL Interconnect: Shared Memory and Storage

GPU-initiated KV access



CXL
switch nodes

Panjiu
server nodes

JBOM nodes
AliMemory /
AliSCM / AliFlash

CXL switch interconnect

Shared pool: 10B IOPS target

CXL Switch Nodes: Roadmap



Evolve from CXL 2.0 to 4.0
Enable Gen7 high-IOPS pooling

CXL Memory Enclosure (JBOM)



Modular AliFlash CXL expansion
Aggregate capacity and IOPS across devices

CXL Compute Head Node



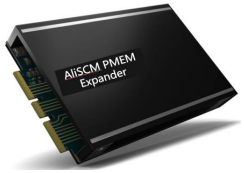
Serve GPU-initiated KV requests
Share the pool across compute nodes

AliMemory



CXL DRAM expansion
Low-latency hot data

AliSCM



CXL PCM persistent memory
Large-capacity warm KV cache

AliFlash CXL SSD



PCIe Gen7 / CXL 4.0 (target)
100M IOPS per device (512 B)

UMX Shared Storage Pool

AliFlash CXL: Device and Pool Targets

Random-read target / device	Useful read bandwidth / device	Example pool size	Aggregate shared-pool target	Memory-semantic access
100M IOPS	**51.2 GB/s**	**100 devices**	**10B IOPS**	**CXL.mem**

Handle fine-grained GPU KV cache access at scale

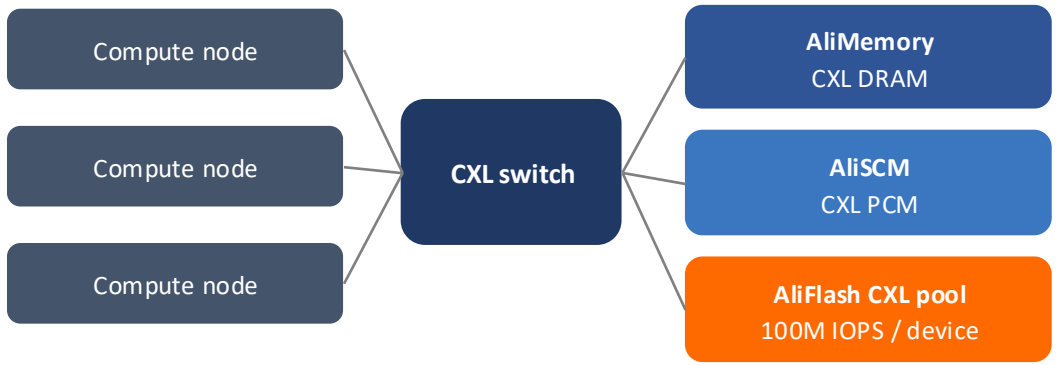
① Read KV data on demand
512 B service granularity reduces unnecessary data movement

② Unified tiering with UMX
HBM / DRAM retain active KV; AliSCM and AliFlash expand shared capacity

③ Share across compute nodes
CXL switches pool AliFlash IOPS and capacity for on-demand allocation

Metric	AliMemory	AliSCM	AliFlash CXL Gen6	AliFlash CXL Gen7 target
Interface / media	CXL / DRAM	CXL / PCM	CXL 3.2 / XL-Flash, pSLC	CXL 4.0 / XL-Flash, pSLC
Read latency	140 ns (64 B)	350 ns (128 B)	10–30 µs	10–30 µs (E2E target)
Random reads	60M IOPS (64 B)	40M IOPS (128 B)	20M IOPS (512 B)	100M IOPS (512 B)
Capacity / module	256 GB	256 GB–1 TB	4 TB	By use case
Primary role	Low-latency hot	Persistent warm	High-capacity warm	High-capacity warm

CXL switch links multiple compute nodes to a tiered resource pool



KV Cache Offload to AliSCM: Measured Gains

Iso-cost L2 comparison: AliSCM 128 GB replaces DRAM 32 GB

Two-layer cache: HBM + L2

30 concurrent users × 30 turns · 500 in / 500 out tokens · ~126 GB KV

Throughput (req/s)



P90 TTFT (s)

DRAM 32 GB	**86.31**	
AliSCM 128 GB	**44.24**	**–48.7%**

Three-layer cache: HBM + L2 + SSD

90 concurrent users × 25 turns · 1024 in / 256 out tokens · ~400 GB KV

Throughput (req/s)



P90 TTFT (s)

DRAM 32 GB	**233.18**	
AliSCM 128 GB	**149.31**	**–36.0%**

Hardware: NVIDIA RTX PRO 5000 · 48 GB L1 GPU memory · PCIe 5.0; host: 8-channel DDR5-6400, 512 GB/socket
Storage: AliSCM CXL 2.0 ×8, 2 × 256 GB interleaved, 21/7 GB/s read/write; L3 SSD: 10/2 GB/s read/write
Inference: SGLang v0.5.6 · Qwen3-8B · multi-turn conversation KV cache test
Workloads: two-layer, 30 users × 30 turns, 500/500 input/output tokens per turn; three-layer, 90 × 25, 1024/256 tokens

Sparse Attention and GPU-Direct Memory

Sparse attention cuts compute and I/O; full primary KV capacity is still required

HBM-resident capacity

1× → 0.1× (example)

HBM holds only the active working set; the full primary KV cache resides in an external warm tier

Small random accesses

~584 B / token / layer

Contiguous runs of ~1.34 tokens/block; weak spatial locality

Serial critical path

0% prefetch benefit assumed

Indexer, TopK selection and Gather execute serially, layer by layer; access latency enters the critical path

External memory needs: capacity, access efficiency and cost

Independent capacity scaling Store the full KV cache	Low cost per GB Support large warm-KV capacity	Small-block high IOPS Serve random row accesses	Low access latency Reduce critical-path stalls

Next-generation designs attach the warm-KV tier to the GPU fabric for capacity scaling and efficient random access

Why UALink for the Scale-Up Fabric

High-performance scale-up interconnect for AI clusters

1

Ultra-high bandwidth

UALink 1.0: 200 Gb/s per lane. Up to 800 Gb/s (100 GB/s) of link bandwidth per port in each direction.

2

Ultra-low latency

Native load/store and atomics. Example switch target: <200 ns port-to-port (50T class, 400 Gb/s ports).

3

Large scale-up domain

UALink 1.0 connects up to 1,024 accelerators across compute nodes within one scale-up pod.

4

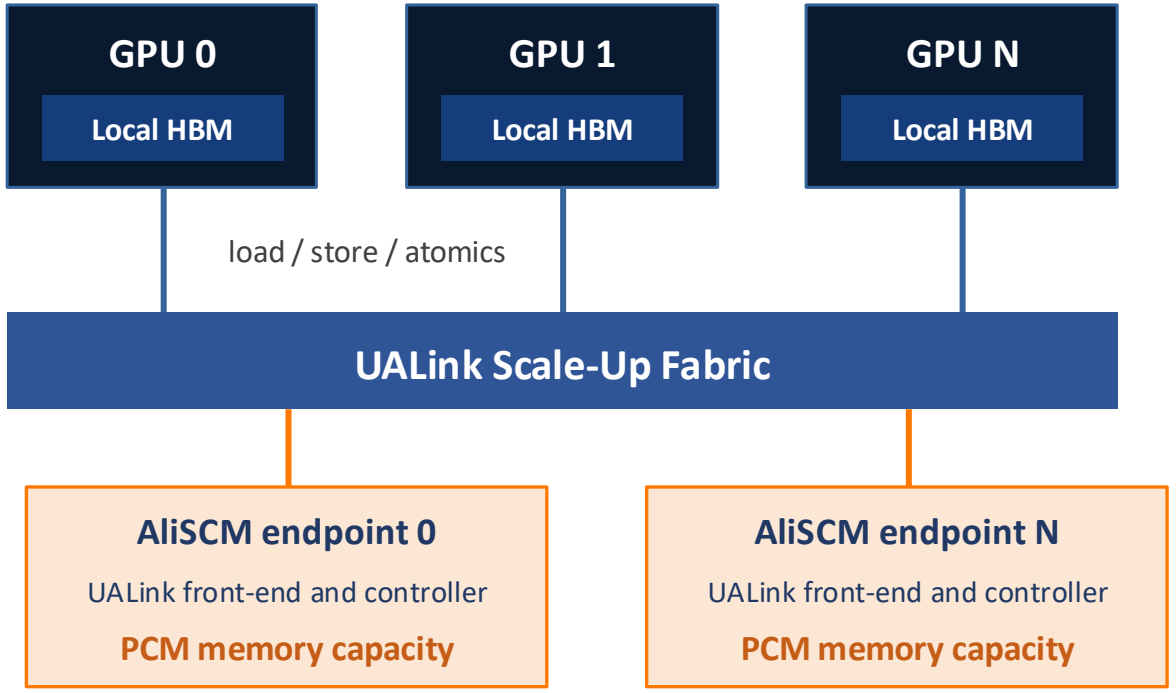
Open ecosystem

An open consortium standard for multi-vendor accelerators and switches, with native memory-semantic communication.

Scale up across compute nodes: connect accelerators through a fabric with native memory semantics

UALink AliSCM: A GPU-Direct PCM Memory Pool

UALink provides native memory access; PCM extends capacity beyond HBM



Native access and address mapping

GPUs issue load/store and atomic operations

Software maps addresses; requests reach memory endpoints directly

HBM and SCM tiering

HBM retains the active working set

PCM holds large warm-KV datasets

Scale capacity independently at lower cost

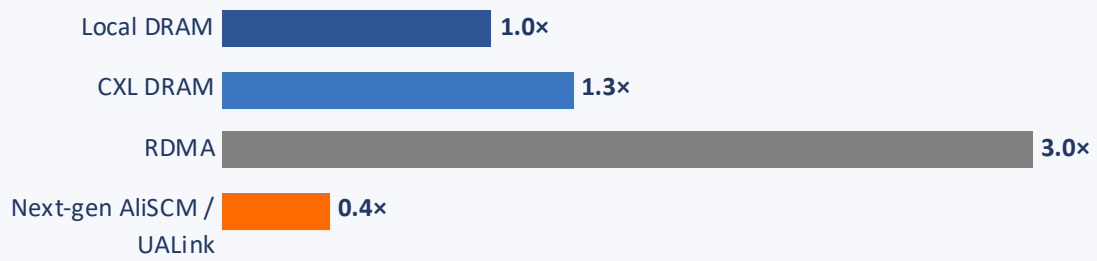
Lower PCM cost per GB

A larger memory pool at the same budget

UALink provides access, AliSCM supplies PCM capacity, and UMX manages resources and tiering

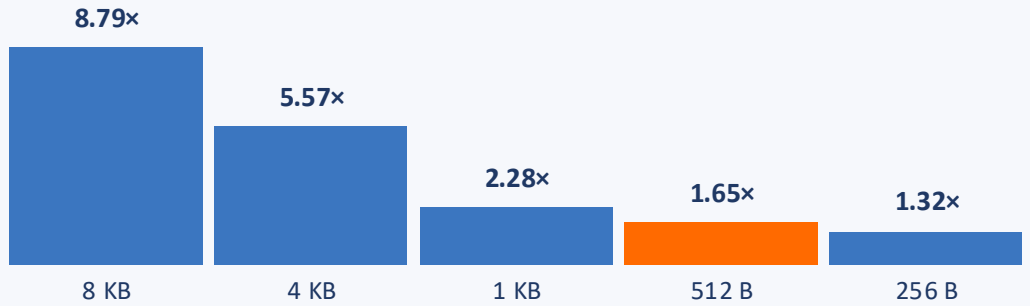
Next-Gen AliSCM via UALink: Estimates

① Estimated latency (Local DRAM = 1)



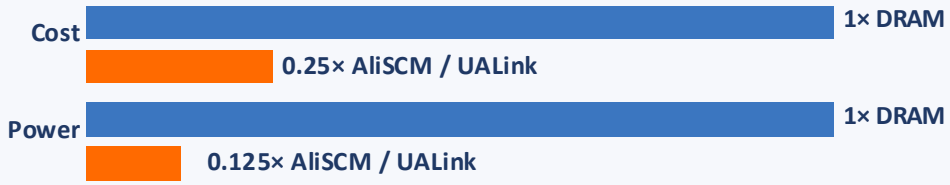
Original design estimate; baseline access path awaits validation

② Read amplification (useful data = 1×)



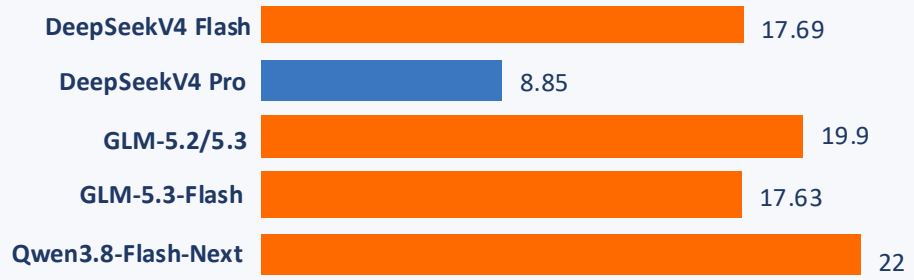
512 B pages balance read amplification and IOPS for 584 B / token / layer

③ Cost and power (DRAM = 1×)



At equal warm-KV capacity, PCM costs ~1/4 as much as DRAM

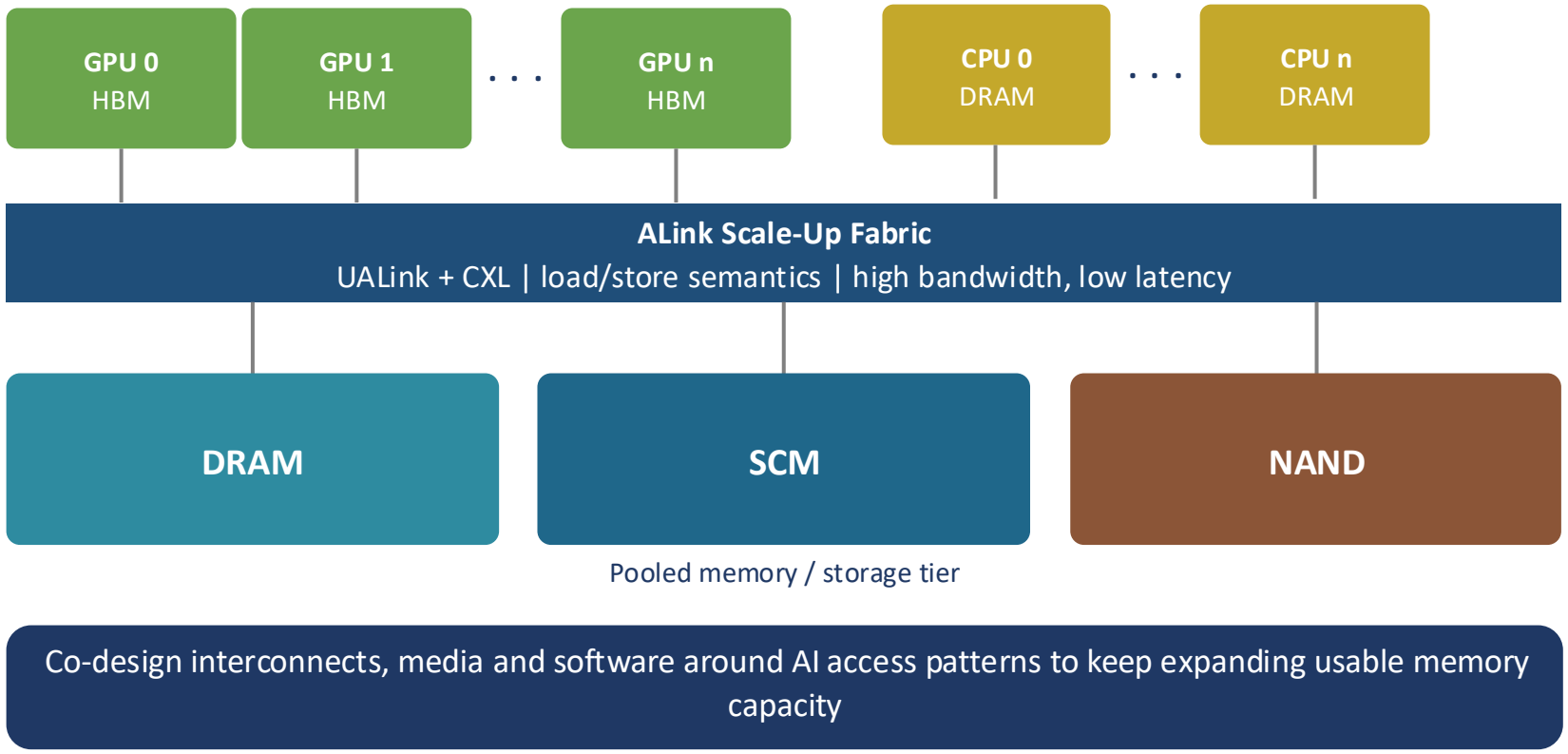
④ Capacity/bandwidth demand: open-source sparse LLMs



AliSCM: 2 TiB / 100 GB/s ≈ 20; capacity/bandwidth matches demand

Long-context KV forms a warm dataset with low activity, served by persistent memory in the scale-up domain

Panjiu UMX: AI Bus-Level Interconnect



Co-design interconnects, media and software around AI access patterns to keep expanding usable memory capacity

Media innovation: persistent-memory tiers

AliSCM and AliFlash CXL SSD form a poolable capacity tier

Application value: KV cache acceleration

Iso-cost two-layer cache
Throughput +58.3%
P90 TTFT −48.7%

Architecture evolution: GPU-direct pools

Native UALink load/store supports sparse attention and long contexts

Thank You!

